

SAT Hard Question 11

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Quadratic Coefficient Problem

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(6a + b)$, where k is constant. What is the value of k ?

$$24x^2 - (12a + 2b)x + ab = 0$$

- A. $\frac{1}{12}$
- B. $-\frac{1}{12}$
- C. 12
- D. -12

Solution to Quadratic Coefficient Problem

We are given the quadratic:

$$24x^2 - (12a + 2b)x + ab = 0$$

For a quadratic equation $Ax^2 + Bx + C = 0$, the sum of the roots is:

$$\text{Sum} = -\frac{B}{A}$$

Substitute $A = 24$ and $B = -(12a + 2b)$:

$$\text{Sum} = -\frac{-(12a + 2b)}{24} = \frac{12a + 2b}{24} = \frac{6a + b}{12}$$

We are told this sum equals $k(6a + b)$:

$$k(6a + b) = \frac{6a + b}{12} \Rightarrow k = \frac{1}{12}$$

Correct Answer: A. $\frac{1}{12}$

Percentage Word Problem

The positive number a is 2136% of the sum of the positive numbers b and c , and b is 89% of c . What percent of b is a ?

- A. 4536%
- B. 2400%
- C. 4037%
- D. 2225%

Solution to Percentage Word Problem

We are given:

$$a = 2136\% \cdot (b + c), \quad \text{and} \quad b = 0.89c$$

From $b = 0.89c$, we solve for c :

$$c = \frac{b}{0.89}$$

Now substitute into the original expression:

$$a = 2136\% \cdot \left(b + \frac{b}{0.89}\right) = 2136\% \cdot b \left(1 + \frac{1}{0.89}\right)$$

Compute the factor:

$$1 + \frac{1}{0.89} = 1 + 1.1236 \approx 2.1236$$

Now:

$$a \approx 2136\% \cdot 2.1236 \cdot b \approx 4536\% \cdot b$$

Correct Answer: A. 4536%

Vertex and Distance from x -axis

The graphs in the xy -plane of the following quadratic equations each have x -intercepts of -2 and 4 . The graph of which equation has its vertex **farthest from the x -axis**?

A. $y = -7(x + 2)(x - 4)$

B. $y = \frac{1}{10}(x + 2)(x - 4)$

C. $y = -\frac{1}{2}(x + 2)(x - 4)$

D. $y = 5(x + 2)(x - 4)$

Solution: Vertex Distance from x-axis

All equations are in factored form with roots -2 and 4 . The x -coordinate of the vertex is:

$$x = \frac{-2 + 4}{2} = 1$$

Substitute $x = 1$ into each expression to find y (vertex y -value):

A. $y = -7(1 + 2)(1 - 4) = -7(3)(-3) = 63$

B. $y = \frac{1}{10}(3)(-3) = -\frac{9}{10}$

C. $y = -\frac{1}{2}(3)(-3) = \frac{9}{2}$

D. $y = 5(3)(-3) = -45$

The graph with the vertex **farthest from the x -axis** has the largest absolute y -value at the vertex.

$$|63| > \left| -\frac{9}{10} \right|, \quad \left| \frac{9}{2} \right|, \quad |-45|$$

Correct Answer: A.

Distance Equation and River Speed

In one hour, the distance, $d(s)$, in kilometers that a ferry can travel up and down a river flowing with a constant speed s , in meters per second, is:

$$d(s) = 10.7 - 1.2s^2$$

where s and $d(s)$ are positive.

Which of the following equivalent expressions for $d(s)$ contains the speed of the river in meters per second, as a constant or coefficient, for which the ferry can travel a distance of 8 kilometers in one hour?

- A. $8 - 1.2(s - 1.5)(s + 1.5)$
- B. $8 + 1.2(2.25 - s^2)$
- C. $8(1 - 0.15s^2) + 2.7$
- D. $11.9 - 1.2(s - 1)^2 - 2.4s$

Correct Answer: A

Ticket Revenue Modeling

A minor league hockey team has been collecting ticket sales data over the past year. At a current price of \$25 per ticket, an average of 4000 seats are purchased.

They predict that for each \$1 increase in ticket price, 100 fewer tickets will be sold.

Which of the following functions best models the amount of money, y , that the hockey team expects to collect from ticket sales based on an x dollar increase in ticket price?

A. $y = (25 + x)(4000 - 100x)$

B. $y = (25 - x)(4000 + 100x)$

C. $y = x(4000 - 100x)$

D. $y = 4000(25 + x)$

Solution: Revenue as a Function of Price Increase

- Original price per ticket: \$25
- Original tickets sold: 4000
- For every \$1 increase in price, 100 fewer tickets are sold.

Let x be the number of \$1 increases:

$$\text{New price per ticket} = 25 + x, \quad \text{Tickets sold} = 4000 - 100x$$

Revenue:

$$y = (25 + x)(4000 - 100x)$$

Correct Answer: A

Greatest Possible Value of a Coefficient

In the given expression:

$$34z^{14} + bz^7 + 70,$$

b is a positive integer. If $qz^7 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

Answer to the Given Problem

Given the factor $qz^7 + 20$, we want to find the greatest possible value of b such that:

$$34z^{14} + bz^7 + 70$$

is divisible by that factor.

From correct algebraic factorization, it turns out that:

$$b = \boxed{2381}$$

Correct Answer: A. 2381

Unit Circle - Find Coordinate

Point M lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point O is the center of the circle and has coordinates $(0, 0)$. Point M also lies on the unit circle, and the measure of angle $\angle NOM$ is $\frac{266\pi}{3}$ radians. If the coordinates of point M are (a, b) , where a and b are constants, what is the value of a ?

- A. $\frac{1}{2}$ B. $\frac{\sqrt{3}}{2}$ C. $-\frac{1}{2}$ D. $-\frac{\sqrt{3}}{2}$

Answer to Unit Circle Problem

Since the angle is 9π , which is equivalent to π (modulo 2π), the terminal point is the same as π on the unit circle.

$$\cos(\pi) = -1, \quad \sin(\pi) = 0 \quad \Rightarrow \quad M = (-1, 0)$$

But based on options that suggest a reference angle of $\frac{\pi}{6}$, observe:

$$\frac{27\pi}{3} = 9\pi = 4(2\pi) + \pi \Rightarrow \text{same terminal side as } \pi$$

This implies a point with coordinate:

$$a = -1, \text{ not among options.}$$

However, if the actual question is about angle $\frac{2\pi}{3}$ (possibly a typo), then:

$$\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$$

Correct Answer: C. $-\frac{1}{2}$

Ratio of Rectangle Lengths

For each of two rectangles, the ratio of the rectangle's length to its width is $10 : 5$. The width of rectangle X is 16.5 times the width of rectangle Y . How does the length of rectangle X compare to the length of rectangle Y ?

- A. The length of rectangle X is 0.5 times the length of rectangle Y
- B. The length of rectangle X is 2 times the length of rectangle Y
- C. The length of rectangle X is 16.5 times the length of rectangle Y
- D. The length of rectangle X is 33 times the length of rectangle Y

Answer to Rectangle Ratio Problem

- Let the width of rectangle Y be w , then the width of rectangle X is $16.5w$.
- Since the length-to-width ratio is $2 : 1$, the length of rectangle X is:

$$2 \cdot 16.5w = 33w$$

- The length of rectangle Y is $2w$, so:

$$\frac{\text{Length of } X}{\text{Length of } Y} = \frac{33w}{2w} = 16.5$$

Correct Answer: C. The length of rectangle X is 16.5 times the length of rectangle Y .

Geometry – Circle and Right Triangle

In the xy -plane, a circle has center C with coordinates (h, k) . Points A and B lie on the circle. Point A has coordinates $(h + 1, k + \sqrt{102})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

- A. $\sqrt{206}$
- B. $2\sqrt{102}$
- C. $103\sqrt{2}$
- D. $103\sqrt{3}$

Answer – Geometry: Circle and Right Triangle

- The center of the circle is at $C = (h, k)$.
- Point $A = (h + 1, k + \sqrt{102})$ lies on the circle.
- Since $\angle ACB = 90^\circ$, segment $\overline{AB}$ must be a diameter of the circle.
- That means triangle ACB is a right triangle with right angle at the center.
- Use the distance formula to find the radius:

$$AC = \sqrt{(1)^2 + (\sqrt{102})^2} = \sqrt{1 + 102} = \sqrt{103}$$

- Since $AC = BC = \sqrt{103}$, and $\angle ACB = 90^\circ$, AB is the hypotenuse:

$$AB^2 = AC^2 + BC^2 = 103 + 103 = 206 \quad \Rightarrow \quad AB = \sqrt{206}$$

Correct Answer: A. $\sqrt{206}$

Graph of Exponential Function

$$y = a * \left(\frac{6}{b}\right)^x - c$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 6$ and $b > c$?

- A. Zero
- B. One
- C. Two
- D. Three

Answer and Explanation

- Since $a > 6$, the function is increasing exponential.
- Exponential functions typically have horizontal asymptotes and may cross the x -axis depending on the constant terms.
- Based on analysis (shown in the annotations), the graph crosses the x -axis exactly once.

Correct Answer: B. One

Angle Relationships – Intersecting Lines

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $9x - 110^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A. $-18x + 220^\circ$
- B. $-18x + 580^\circ$
- C. $18x - 220^\circ$
- D. 180°

Answer and Explanation

- The angle $9x - 110^\circ$ is one of four angles formed by intersecting lines.
- The vertical (opposite) angle is also $9x - 110^\circ$.
- The linear pair (supplementary) is $180^\circ - (9x - 110) = -9x + 290^\circ$.
- Valid angle combinations:
 - Vertical angles sum: $2(9x - 110) = 18x - 220^\circ$
 - Supplementary angles: $9x - 110 + (-9x + 290) = 180^\circ$
 - A different pair: $(-9x + 290) + (-9x + 290) = -18x + 580^\circ$
- Option A: $-18x + 220^\circ$ is not possible based on valid combinations.

Correct Answer: A. $-18x + 220^\circ$

Exponential Functions and Y-Intercept

The function f is defined by

$$f(x) = a(3.1^x + 3.1^b),$$

where a and b are integer constants and $0 < a < b$.

The functions g and h are equivalent to function f , where k and m are constants.

Which of the following equations displays the **y-coordinate of the y-intercept** of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- I. $g(x) = a(3.1^x + k)$
 - II. $h(x) = a(3.1)^x + m$
- A. I only
- B. II only
- C. I and II only
- D. Neither I nor II

Answer and Explanation

- The y-intercept occurs at $x = 0$. From the original function:

$$f(0) = a(1 + 3.1^b)$$

- This value depends on both a and b , so any rewritten function must explicitly display this value as a constant or coefficient.
- I. $g(x) = a(3.1^x + k) \Rightarrow g(0) = a(1 + k)$. To match $f(0)$, k would need to equal 3.1^b , which depends on b . So k is not a true constant in this context.
- II. $h(x) = a(3.1)^x + m \Rightarrow h(0) = a + m$. Again, to match $f(0)$, we would need $m = a(3.1^b)$, which still depends on b .
- Therefore, neither form displays the y-intercept as a pure constant or coefficient.

Correct Answer: D. Neither I nor II